Rmk: Property for arbitary series $\sum a_n$

① $\sum a_n$ 收敛 $\Rightarrow \lim\limits_{n\to\infty} a_n = 0$

② 收敛±发散=发散；收敛±收敛=收敛

③ $\sum|a_n|$收敛，即$\sum a_n$绝对收敛 $\Rightarrow$ $\sum a_n$ 收敛.

Test only for series $\sum a_n$ with $a_n \geq 0$

① Integral test: $\sum f(n)$ & $\int_N^\infty f(x)dx$ 同敛散

② comparsion test. (& limit form)

③ ratio test: $\lim\limits_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \rho < 1 \Rightarrow$ 绝对收敛

④ root test. $\lim\limits_{n\to\infty}\sqrt[n]{|a_n|} = \rho < 1 \Rightarrow$ 绝对收敛

Test for Alternating series

$\sum (-1)^{n+1} u_n$ 收敛若

(i) $u_n \geq 0$

(ii) u_n 单调递减

(iii) $u_n \xrightarrow{n\to\infty} 0$

Rmk: ① product rule for limit $(-1)^n \cdot \frac{(-1)^n}{n}$

② bounded $\not\Rightarrow$ limit exist. $(-1)^n$

补充作业 2

Q1. P24-25 Ex 1, 2, 4, 9, 10

1. Assume $0 < a_n < \frac{1}{n}$, which of the following series converges for sure?

(A) $\sum_{n=1}^{\infty} a_n$.

(B) $\sum_{n=1}^{\infty} (-1)^n a_n$.

(C) $\sum_{n=1}^{\infty} \sqrt{a_n}$.

(D) $\sum_{n=1}^{\infty} \frac{a_n}{n}$.

(B) 反例: $a_n = \frac{1}{4}\left[\frac{2}{n} + \frac{(-1)^n}{n}\right]$

原因: $0 < \frac{a_n}{n} < \frac{1}{n^2}$

D

2. Assume both series $\sum u_n$ and $\sum v_n$ diverge. Then

(A) $\sum_{n=1}^{\infty}(u_n + v_n)$ diverges.

(B) $\sum_{n=1}^{\infty} u_n v_n$ diverges.

(C) $\sum_{n=1}^{\infty}(|u_n| + |v_n|)$ diverges.

(D) $\sum_{n=1}^{\infty}(u_n^2 + v_n^2)$ diverges.

原因: $|u_n| + |v_n| \geq |u_n|$

C

4. Suppose a is a real number. Then the series $\sum_{n=1}^{\infty}\left(\frac{\sin a}{n^2} - \frac{1}{\sqrt{n}}\right)$

(A) converges absolutely.

(B) converges conditionally.

(C) diverges.

(D) convergence depends on a.

C

9. If $\sum_{n=1}^{\infty}(-1)^{n-1}u_n = 2$ and $\sum_{n=1}^{\infty} u_{2n-1} = 5$, then $\sum_{n=1}^{\infty} u_n =$ (8).

10. Assume $\sum_{n=2}^{\infty} \frac{1}{n \ln^p n}$ converges, the range of p is ($p>1$).

Q2.

2. Which of the following series converge absolutely, which converge conditionally, and which diverge ? Give your reasons.

(a) $\sum_{n=1}^{\infty}(-1)^n 2^n \sin\frac{\pi}{3^n}$. (b) $\sum_{n=1}^{\infty}\left(\frac{\sin n}{n^2}+\frac{(-1)^n}{n}\right)$. (c) $\sum_{n=1}^{\infty}\frac{n!}{n^n}$. (d) $\sum_{n=1}^{\infty}\frac{(\ln n)^2}{n}$.

Sol: (a) 绝对收敛，因为

$$\lim_{n\to\infty}\frac{2^n\sin\frac{\pi}{3^n}}{\left(\frac{2}{3}\right)^n}=\pi$$

$\square$

(b) 条件收敛，因为 $\sum\frac{|\sin n|}{n^2}\le\sum\frac{1}{n^2}<\infty$，$\sum\frac{(-1)^n}{n}<\infty$

并且 $\left|\frac{\sin n+(-1)^n n}{n^2}\right|=\begin{cases}\frac{n-\sin n}{n^2} & n\text{ 奇}\\ \frac{n+\sin n}{n^2} & n\text{ 偶}\end{cases}$

$$=\frac{n+(-1)^n\sin n}{n^2}$$

$$=\frac{1}{n}+\frac{(-1)^n\sin n}{n^2}$$

还有 $\sum\frac{1}{n}$ 发散，$\sum\frac{(-1)^n\sin n}{n^2}$ 收敛

$\therefore\ \sum\left|\frac{\sin n+(-1)^n n}{n^2}\right|$ 发散.

$\square$

(c) 绝对收敛，因为比较判别法

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\frac{(n+1)!}{(n+1)^{n+1}}\frac{n^n}{n!}=\lim_{n\to\infty}\left(\frac{n}{n+1}\right)^n=e^{-1}<1$$

$\square$

(d) 发散

因为 $\frac{(\ln n)^2}{n} > \frac{1}{n}$ 对 $n >> 0$，而 $\sum \frac{1}{n}$ 发散

$\Longrightarrow \sum \frac{(\ln n)^2}{n}$ 发散

Q3 P25 Ex19

19. Find a, b, c such that

$$\lim_{n\to\infty} n(an + \sqrt{2 + bn + cn^2}) = 2.$$

Sol: 注意到 $a<0$, $c\geq 0$

$$2 = 原式 = \lim_{n\to\infty} \frac{n(2+bn+cn^2-(an)^2)}{\sqrt{2+bn+cn^2} - an}$$

$$= \lim_{n\to\infty} \frac{2+bn+(c-a^2)n^2}{\sqrt{\frac{2}{n^2}+\frac{b}{n}+c} - a} \Rightarrow \begin{cases} c=a^2 \\ b=0 \\ \frac{2}{\sqrt{c}-a}=2 \end{cases} \Rightarrow \begin{cases} a=-\frac{1}{2} \\ b=0 \\ c=a^2=\frac{1}{4} \end{cases}$$

$\square$

Q4 P26 Ex25

25. Determine if the following series converges or not

$$\sum_{n=1}^{\infty}\left(1-\cos\frac{1}{n}\right)^p, \quad p>0.$$

Sol: $\lim_{n\to\infty} \frac{\left|1-\cos\frac{1}{n}\right|^p}{\left|\frac{1}{n^2}\right|^p} = \lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)^p = \frac{1}{2^p}$

$\therefore$ consider $\sum \frac{1}{n^{2p}}$: $\begin{cases} 收敛 & 0<p\leq\frac{1}{2} \\ 发散 & p>\frac{1}{2} \end{cases}$

Q5 P 27 Ex35

35. If $\sum a_n$ converges absolutely, prove that $\sum \frac{a_n}{a_n+1}$ converges absolutely.

$$\text{pf: } \lim_{n\to\infty} \frac{\left|\frac{a_n}{a_n+1}\right|}{|a_n|} = \lim_{n\to\infty} \left|\frac{1}{a_n+1}\right| = 1$$

$\therefore$ $\sum |a_n|$ 收斂 $\Rightarrow$ $\sum \left|\frac{a_n}{a_n+1}\right|$ 收斂.

D